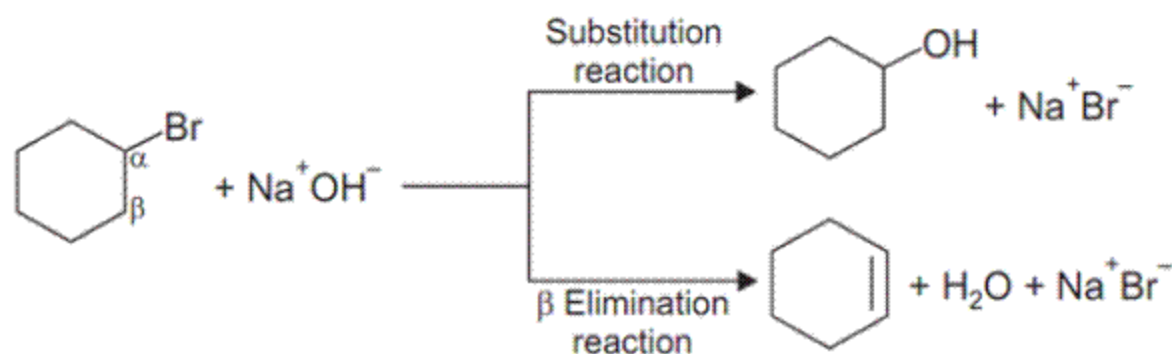
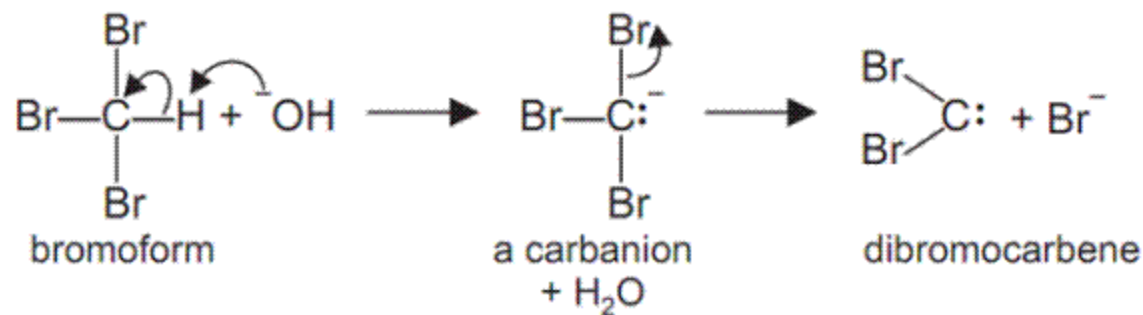
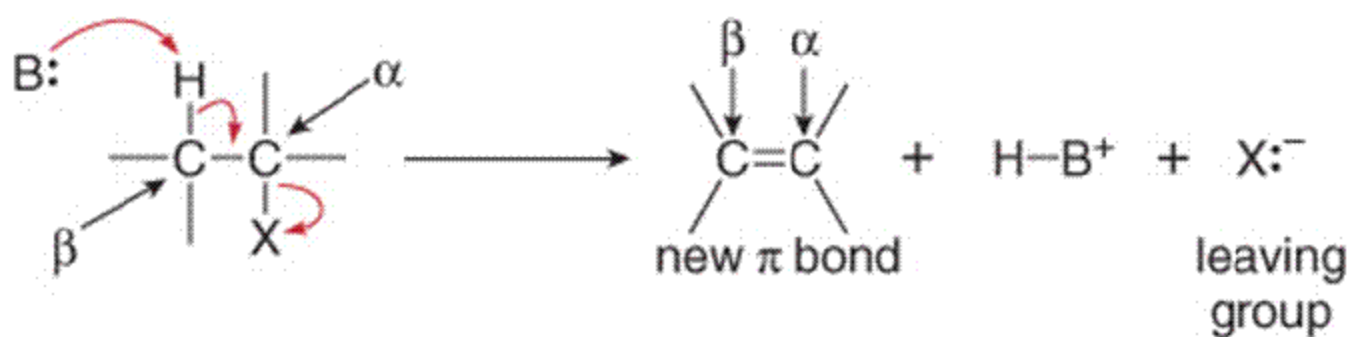
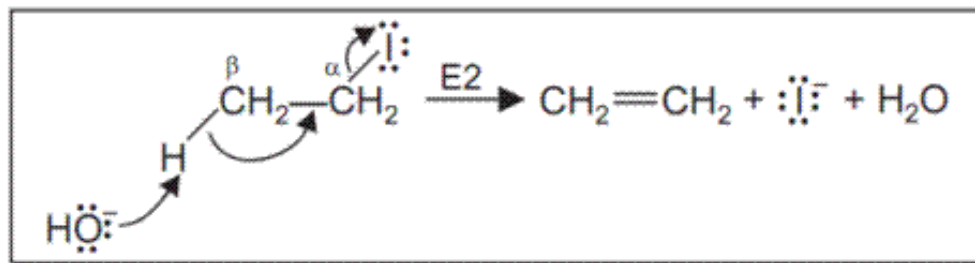
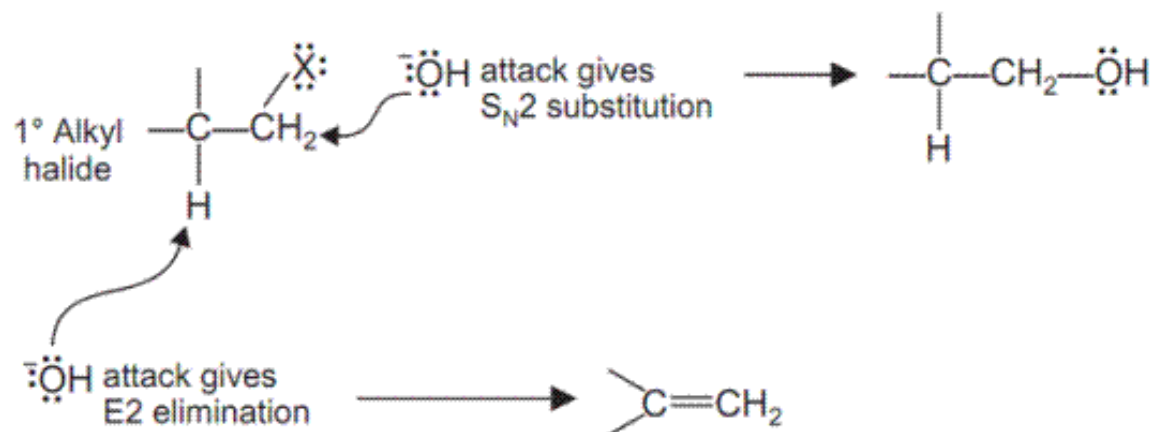
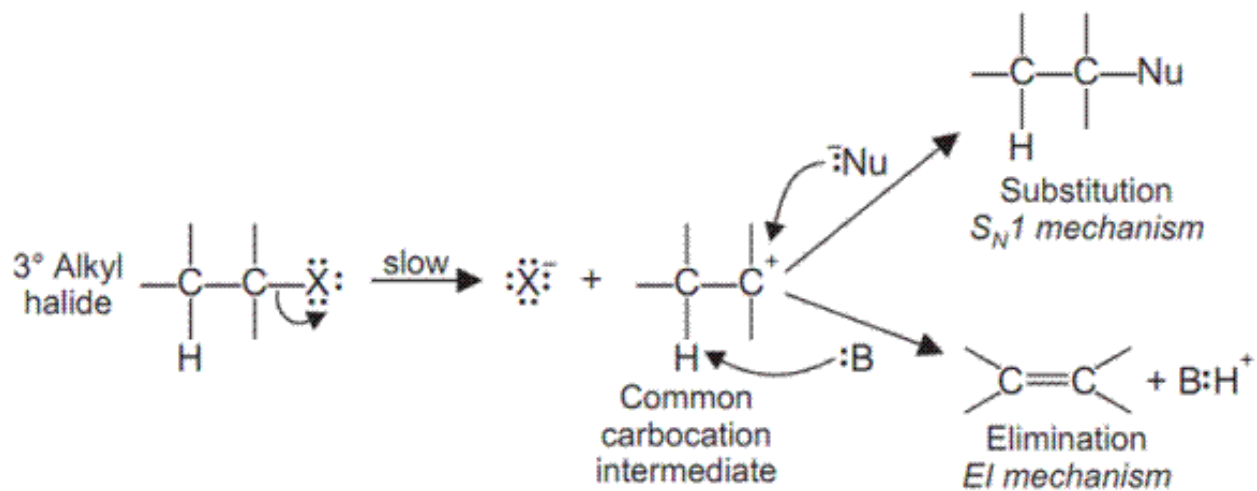
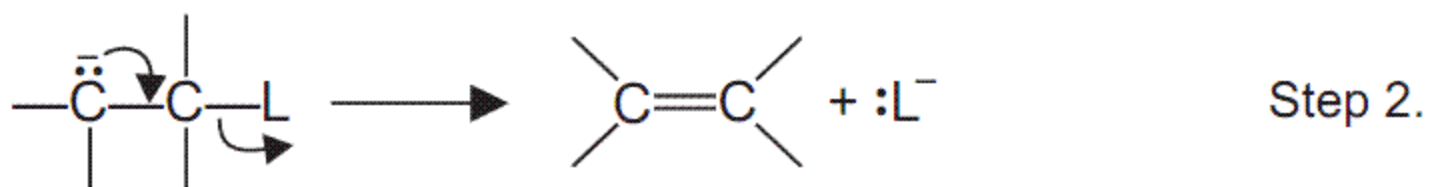
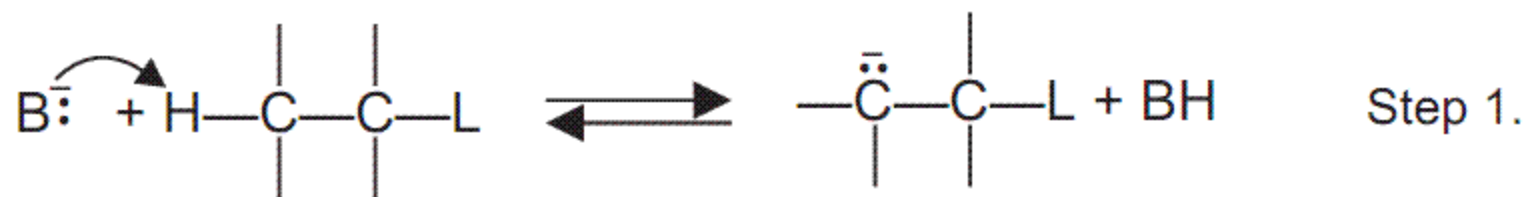




Generally in S_N1 and S_N2 reactions, substitution is accompanied by some elimination of H-L , where L is a leaving group to give an alkene.

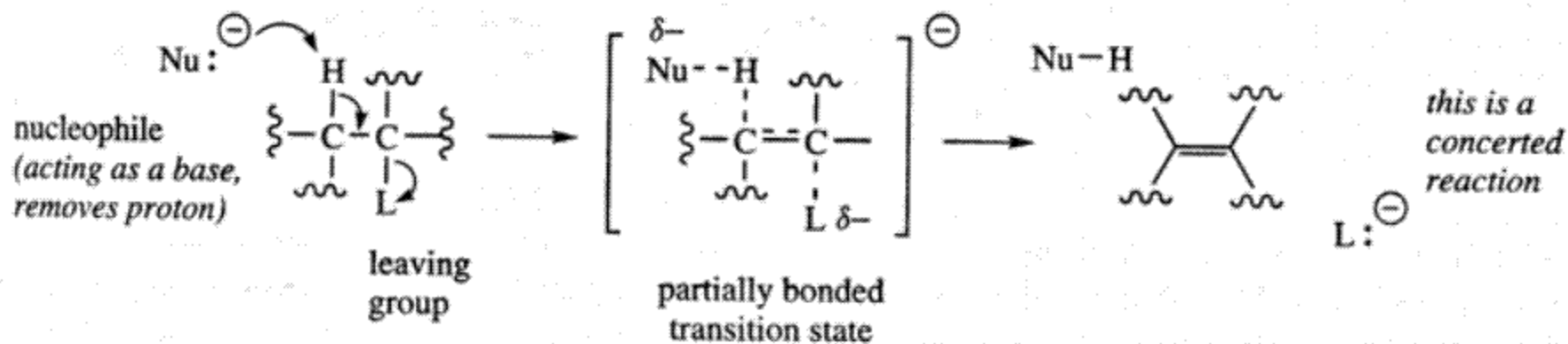




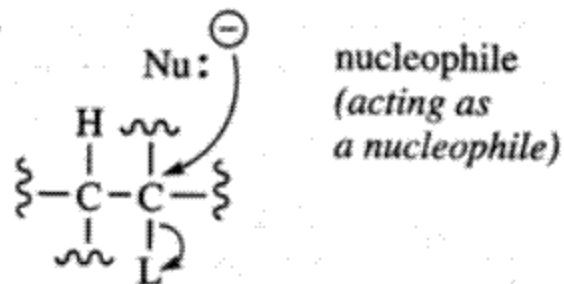


ElcB Mechanism: Carbanion Intermediates

E2 mechanism

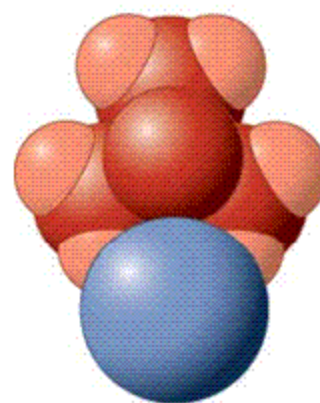
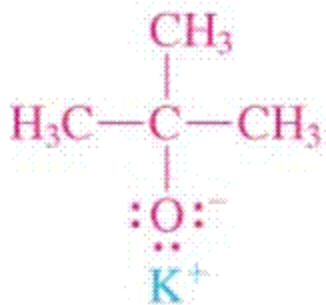


compare an S_N2
mechanism

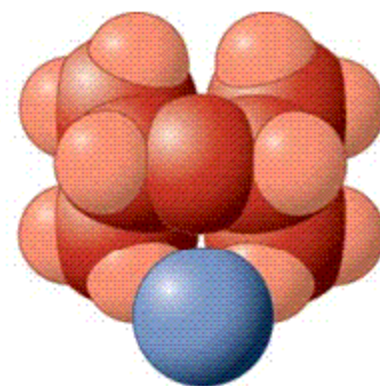




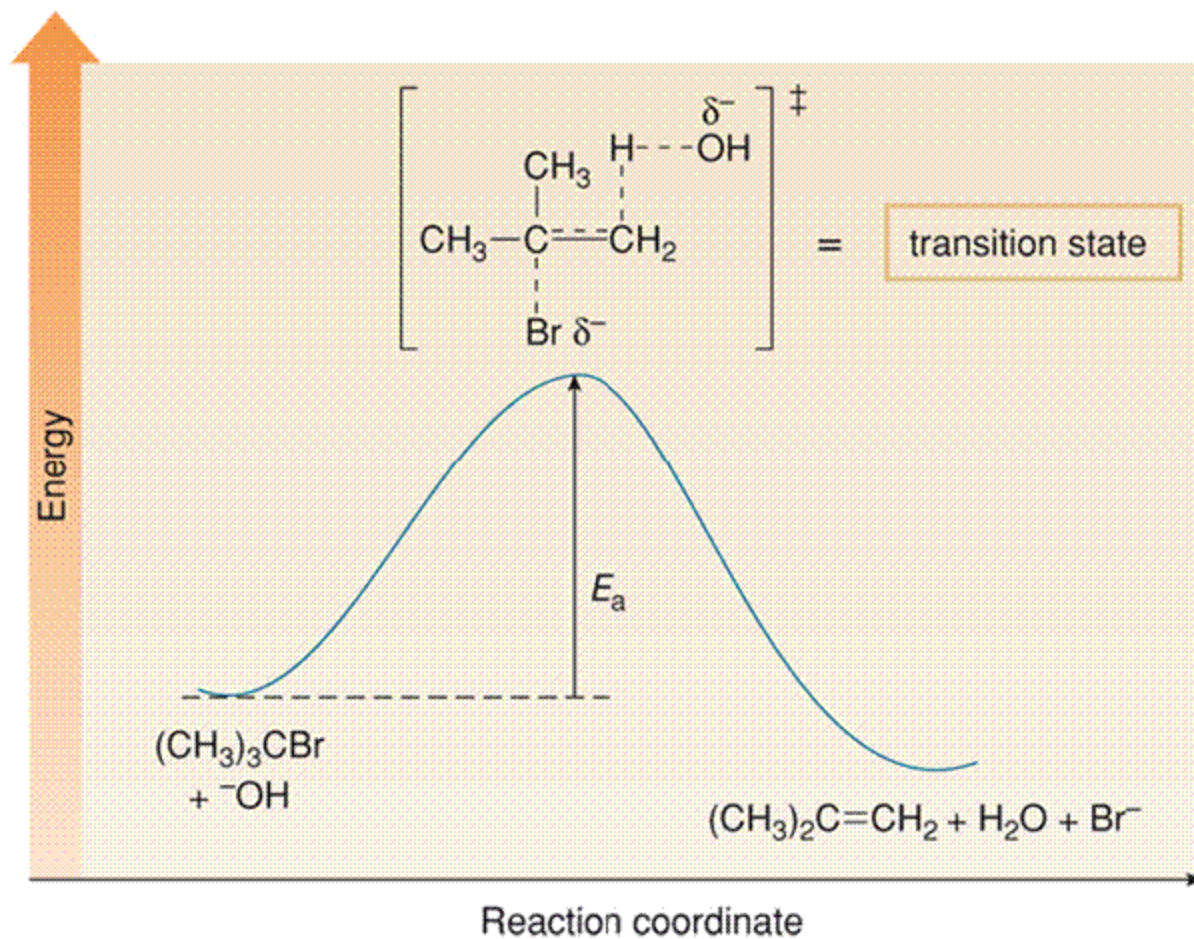
Sterically Hindered Bases



Potassium *tert*-butoxide



Lithium diisopropylamide
(LDA)



- Because the bond to the leaving group is partially broken in the transition state, the better the leaving group the faster the E2 reaction.

Order of reactivity
of RX:

R-F

R-Cl

R-Br

R-I

Increasing leaving group ability
Increasing rate of the E2 reaction

- As the number of R groups on the carbon with the leaving group increases, the rate of the E2 reaction increases.

Increasing rate of an S_N2 reaction

RCH₂-X

R₂CH-X

R₃C-X

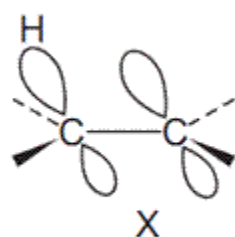
1°

2°

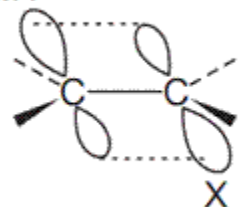
3°

Increasing rate of an E2 reaction

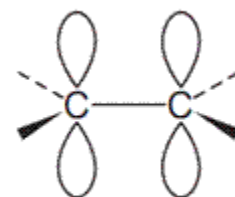
Substituents are anti-periplanar (a staggered conformer)



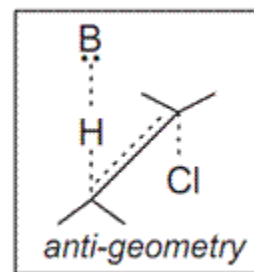
base.....H



anti elimination

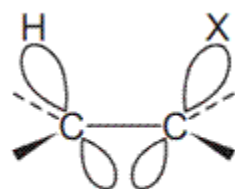


product alkene

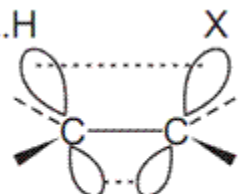


anti-geometry

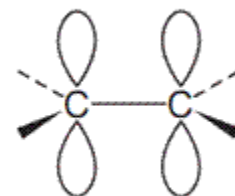
Substituents are syn-periplanar (an eclipsed conformer)



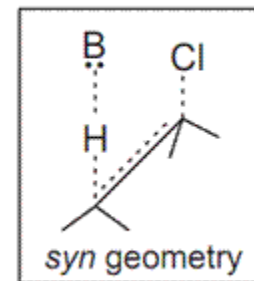
base.....H



syn-elimination



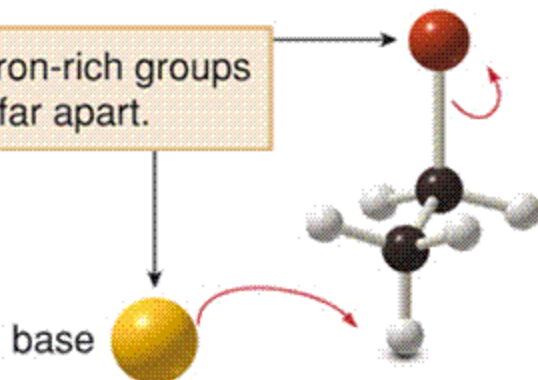
product alkene



syn geometry

An **anti periplanar** arrangement has a **staggered** conformation.

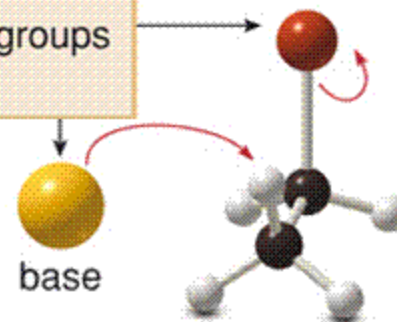
Two electron-rich groups
are far apart.

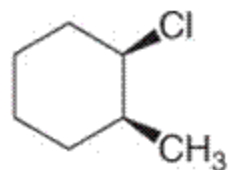


preferred geometry

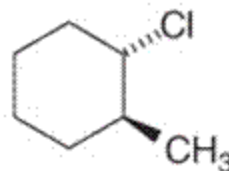
A **syn periplanar** arrangement has an **eclipsed** conformation.

Two electron-rich groups
are close.



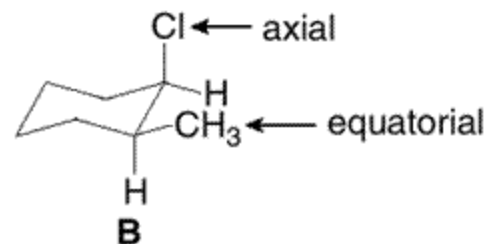
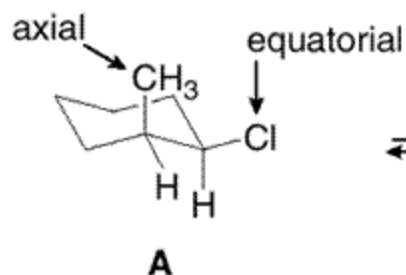


cis-1-chloro-2-methyl-
cyclohexane

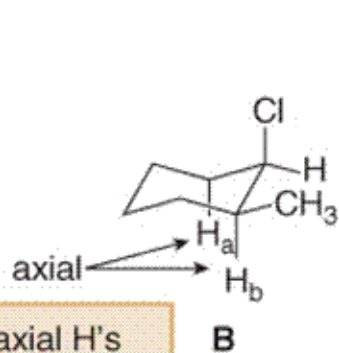


trans-1-chloro-2-methyl-
cyclohexane

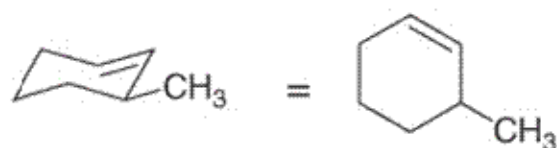
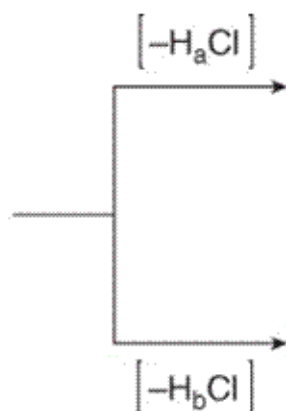
cis isomer



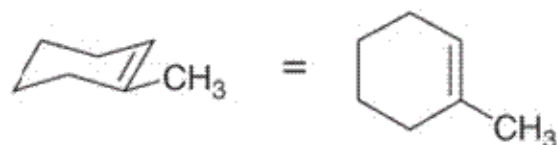
This conformation reacts.



Two β axial H's
Both H's can react.

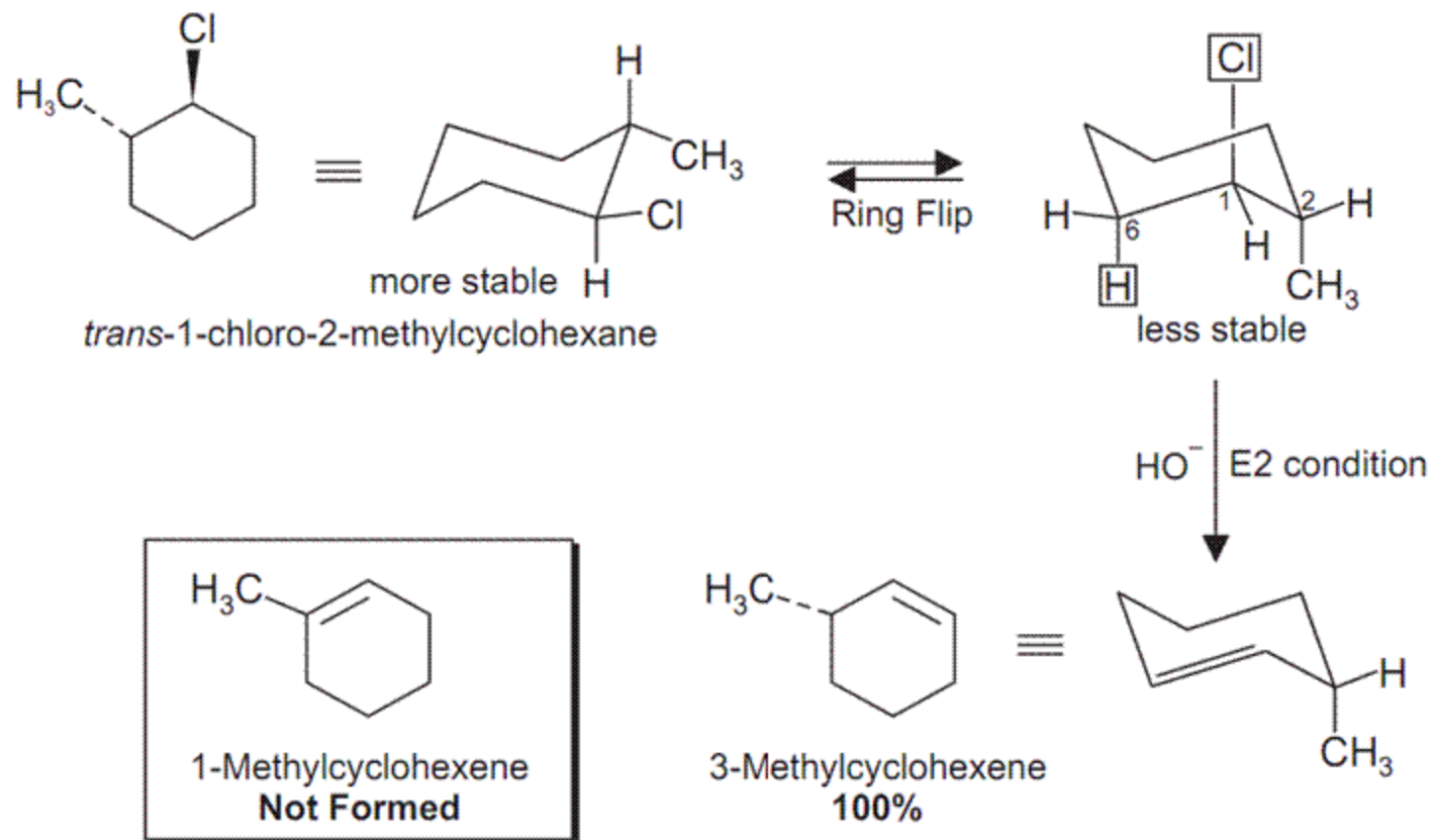


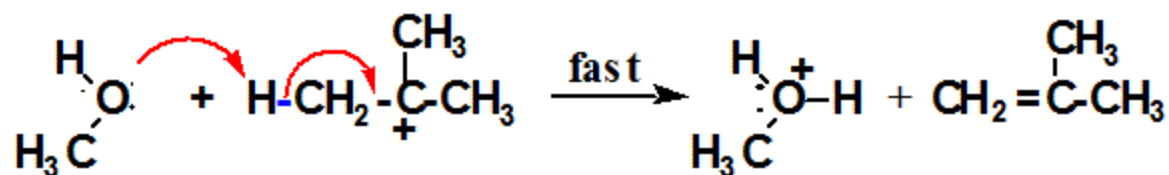
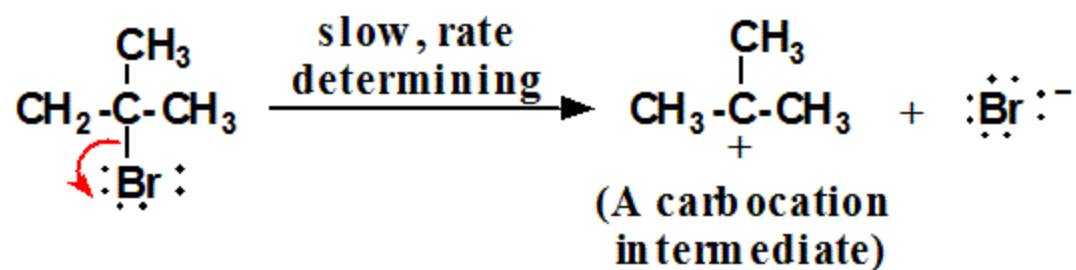
disubstituted alkene
minor product



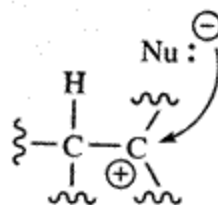
trisubstituted alkene
major product

E2-Elimination from Trans-1-Chloro-2-Methylcyclohexane

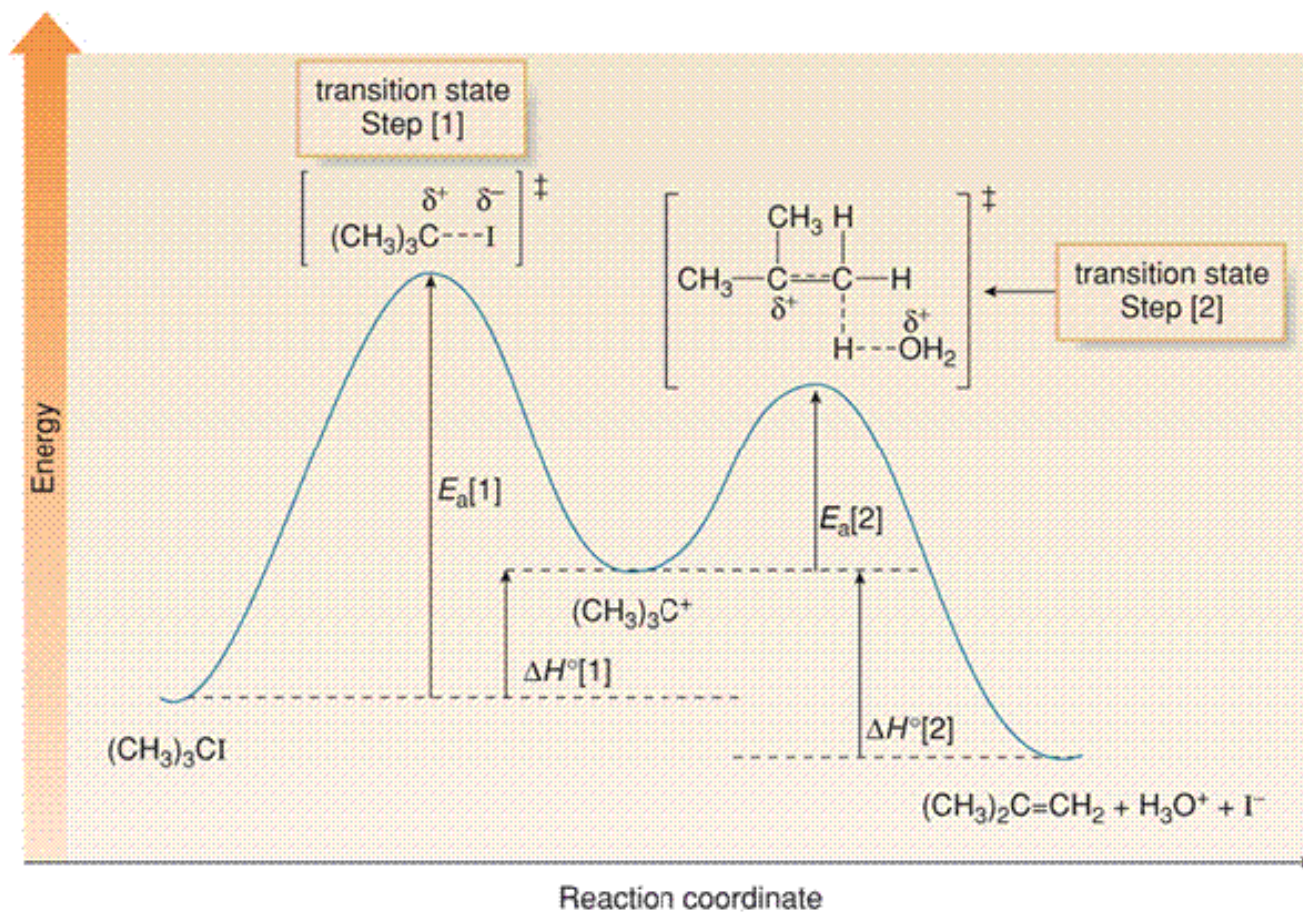


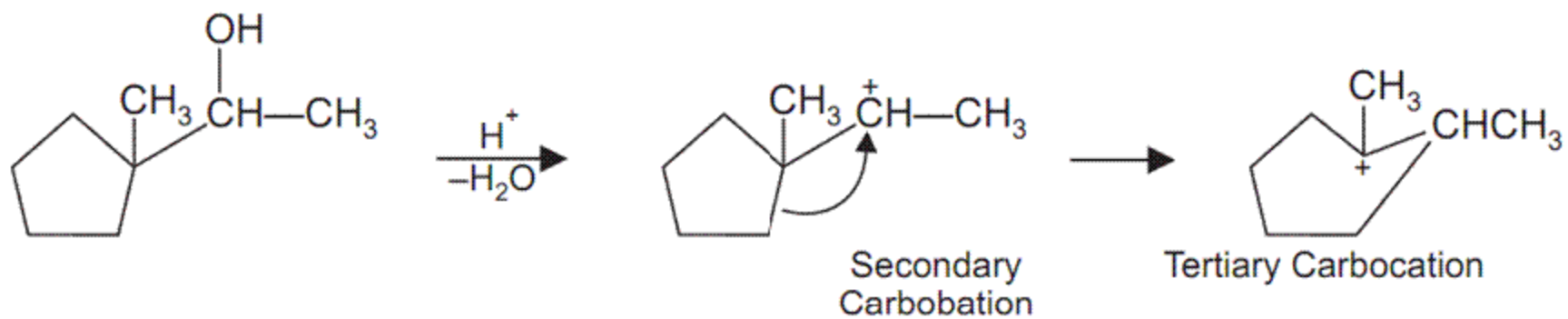


compare an $\text{S}_{\text{N}}1$
mechanism

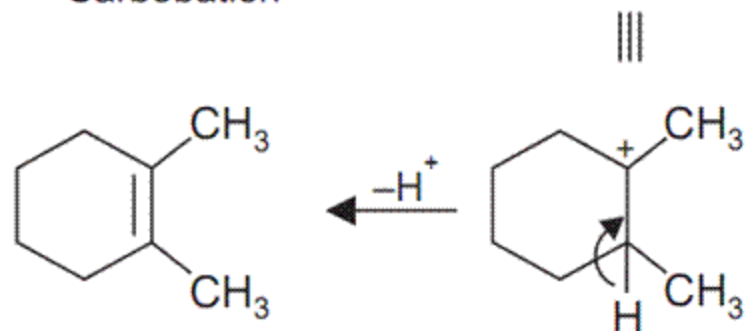


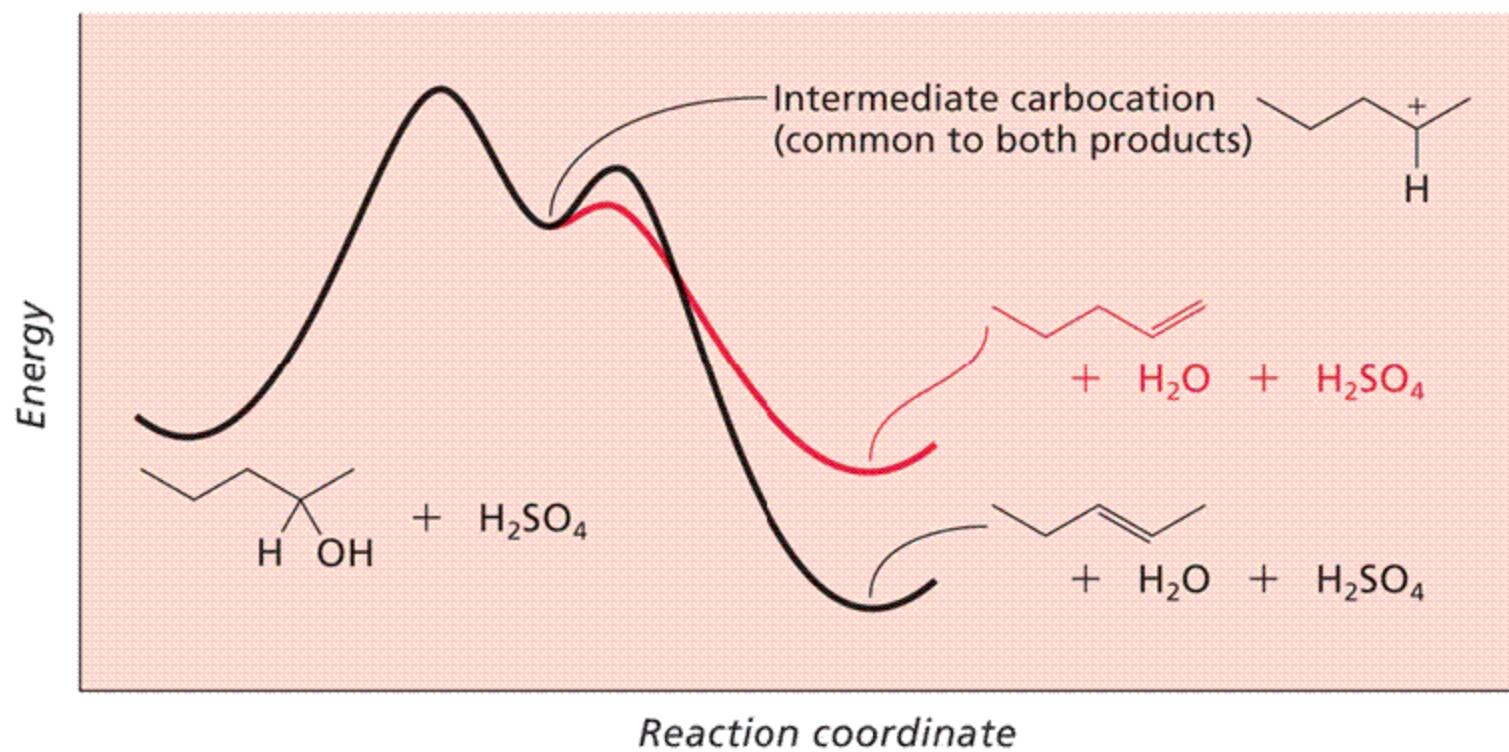
nucleophile
(acting as
a nucleophile)

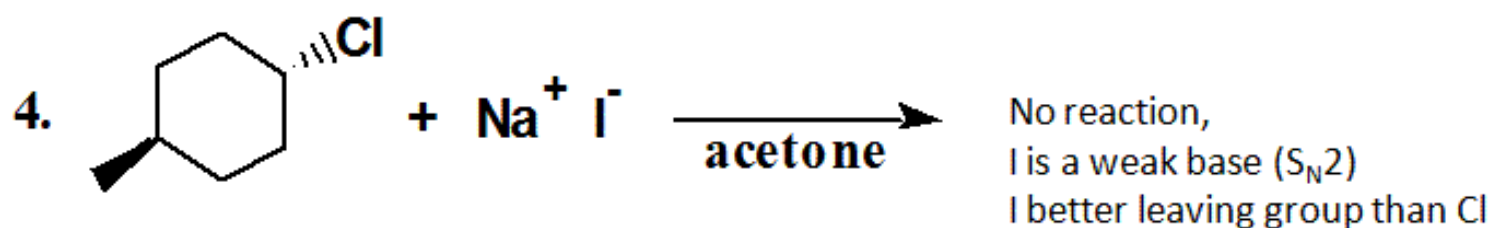
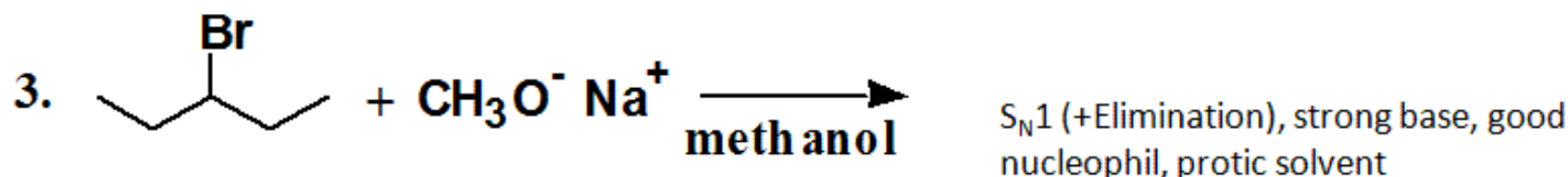
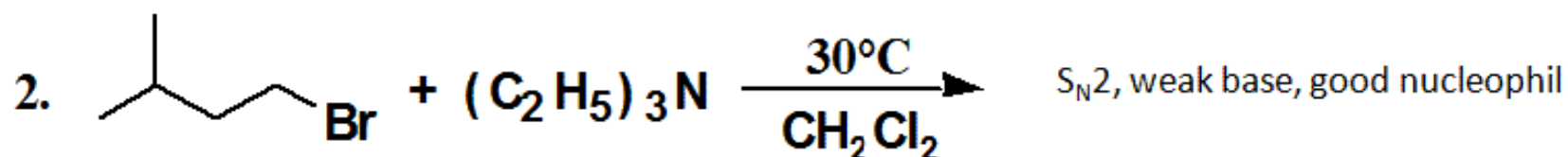
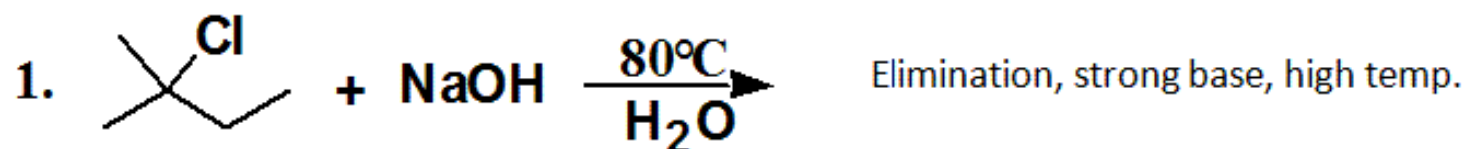




The driving force for the rearrangement is the formation of a more stable carbocation







	S_N1	S_N2	E1	E2
Substrate	Strong effect; reaction favored by tertiary halide	Strong effect; reaction favored by methyl or primary halide	Strong effect; reaction favored by tertiary halide	Strong effect; reaction favored by tertiary halide
Reactivity – primary	Does not occur	Highly favored	Does not occur	Occurs with strong base!
Reactivity – tertiary	Favored when nucleophile is the solvent – solvolysis	Does not occur	Occurs under solvolysis conditions or with strong acids	Highly favored when strong bases (OH ⁻ , OR ⁻) are used
Reactivity – secondary	Can occur in polar, protic solvents	Favored by good nucleophile in polar, aprotic solvents	Can occur in polar, protic solvents	Favored when strong bases are used
Solvent	Very strong effect; reaction favored by polar, protic solvents	Strong effect; reaction favored by polar, aprotic solvents	Very strong effect; reaction favored by polar, protic solvents	Strong effect; reaction favored by polar, aprotic solvent
Nucleophile/Base	Weak effect; reaction favored by good nucleophile/weak base	Strong effect; reaction favored by good nucleophile/weak base	Weak effect; reaction favored by weak base	Strong effect; reaction favored by strong base
Leaving Group	Strong effect; reaction favored by good leaving group	Strong effect; reaction favored by good leaving group	Strong effect; reaction favored by good leaving group	Strong effect; reaction favored by good leaving group